

Coverage Is Not Predictive Fidelity: A Split-Landscape Analysis of Data Condensation for Gradient-Boosted Trees

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Abstract—

Data condensation replaces a large training set with a small surrogate, but tabular condensers are usually judged only after retraining a model on each candidate. For histogram gradient-boosted decision trees (GBDTs), we ask whether a candidate preserves the split decisions the full-data booster would make. We introduce Split-Landscape Discrepancy (SLD), a no-candidate-retraining diagnostic that compares full and condensed split-gain landscapes under fixed preprocessing and prediction state.

Across a broad audit of tabular condensers, method-level split-regret is highly correlated with downstream area under the receiver operating characteristic curve (AUROC) in the small-to-mid-size regime, while a separate core grid supports the headline performance comparisons. SLD is therefore best used as a rejection filter and method-family diagnostic, not as a selector for the single best method on a new dataset. It also exposes why coverage can fail: a greedy high-gain threshold-coverage selector can perform poorly because unweighted coverage differs from gain-weighted structural fidelity. At very large scale, random sampling becomes hard to beat and root-regret loses resolution as reasonable surrogates clear dominant split margins. We release the benchmark, scripts, processed tables, and diagnostic.

Index Terms—data condensation; gradient-boosted decision trees; coreset selection; tabular data; no-candidate-retraining diagnostics

I. INTRODUCTION

ML engineers and data scientists who train gradient-boosted models on tabular data face a recurring operational problem: every candidate condensed dataset is usually judged by retraining a model on it. That is acceptable for a few surrogates, but it becomes costly when condensation is used inside repeated model development, hyperparameter search, data versioning, or governance workflows. A small surrogate can make these workflows cheaper and easier to share, although condensation is not itself a privacy or compliance guarantee. The applied question is therefore not only *which* surrogate gives the best final AUROC, but which candidates are worth retraining on at all.

Existing condensation and coreset methods provide many ways to build such surrogates [1]–[4], including re-

cent methods designed for tabular data [5], [6]. Their common evaluation protocol, however, remains largely black-box: train a fresh model on each condensed set, measure downstream predictive performance, and compare. This is the right final validation step, but it is a poor diagnostic. It does not say whether a bad surrogate failed because it missed important split thresholds, distorted leaf statistics, or simply lacked enough budget. For practitioners, that distinction matters: a useful diagnostic should reject clearly bad surrogates early and explain which failure mode is being observed.

Histogram gradient-boosted decision trees (GBDTs) offer a concrete handle for such a diagnostic. At each node, a GBDT chooses the feature and threshold that maximise a regularised split-gain criterion [7]–[9]. Given a fixed prediction state, the candidate split gains are determined by binned sums of first- and second-order loss statistics. The resulting table of gains is the model’s *split-gain landscape*: it records which structural decisions the booster is likely to make before leaf values are estimated. If a condensed set preserves this landscape, it has a plausible route to preserving the learned tree structure; if it distorts the landscape, downstream retraining is already suspect.

We introduce *Split-Landscape Discrepancy* (SLD), a no-candidate-retraining diagnostic that compares the split-gain landscape of the full data with that of a candidate surrogate under the same preprocessing and fixed prediction state. SLD is not a claim that no model state is ever used; rather, it replaces retraining on every candidate with a histogram-pass screening measurement. In an applied pipeline, its role is to flag high-risk surrogates and compare method families before spending retraining compute on the final candidates.

Using SLD as a measurement instrument, we evaluate 17 condensation methods spanning classical coresets, gradient-based selectors, distribution matching, GBDT-native sampling, and gain-path variants. The broad 17-method audit uses a lighter diagnostic grid, while a fuller five-budget, five-seed core grid supports the headline performance comparisons for the main methods. The study is deliberately not a new condensation algorithm or a state-of-the-art claim. It is a benchmark and diagnostic

analysis: when does a landscape score predict downstream behaviour, when does it fail, and what does that imply for using condensed datasets in GBDT workflows?

Contributions.

- **The SLD diagnostic.** We introduce the Split-Landscape Discrepancy, a no-candidate-retraining metric that quantifies how faithfully a condensed dataset reproduces the GBDT split-gain landscape of the full data under a fixed prediction state (Section III).
- **Systematic comparative analysis.** We present a 17-method diagnostic audit for GBDT-on-tabular settings and a separate five-method core grid over 26 datasets, five budgets, and five seeds, measuring both downstream predictive performance and landscape fidelity (Sections IV and V).
- **Three empirically grounded insights.** A single landscape score is method-level rank-correlated with AUROC at Spearman $\rho = -0.87$ in the small-to-mid-size diagnostic grid, making it useful for screening and family-level ordering rather than top-1 selection. Within-dataset trends are generally negative after controlling for dataset scale. Coverage and predictive fidelity are decoupled: a $(1 - 1/e)$ -optimal coverage selector is among the weakest downstream performers, explained by a reference-chain decomposition into structure and leaf-estimate gaps. Structure is robust to perturbation below a margin-linked threshold; the dose is a fraction of the split margin, not a fraction of intentionally corrupted splits (Section V).
- **Scope boundaries.** We document where the lens does not extend: regression tasks (structure-dominated failure), hyperparameter-rank transfer (near-zero Jaccard), and no-candidate-retraining selection of the single best method (top-3 rate 0.44), measured under the same protocol (Section VI).
- **Reproducibility repository.** The benchmark protocol, SLD implementation, processed tables, seeds, command lines, and plotting scripts are released at <https://github.com/VIVAN-RANGRA/sld-gbdt-condensation>.

II. RELATED WORK

a) Dataset distillation and condensation.: Dataset distillation [1] and its successors optimise a small synthetic set so that a model retrained on it matches one trained on the full data, using gradient matching [2], distribution matching [10], or trajectory matching [11]. Comprehensive surveys [12]–[14] emphasize downstream evaluation, mostly on image benchmarks, and do not provide a GBDT-specific split-landscape fidelity metric. Extending distillation to tabular data has attracted recent attention [5], [15]–[17], but these works retain the retrain-and-evaluate paradigm. Our contribution is orthogonal: rather than a new condensation method, we introduce a no-candidate-retraining diagnostic that measures whether

any condensed set, regardless of the method that produced it, preserves the GBDT split-gain landscape.

b) Coreset selection.: Classical coreset methods select real training examples by geometric or gradient-based criteria: herding [3] matches distribution moments; k-center [18] minimises the maximum coverage distance; the CRAIG coreset method [4] and Gradient Matching (GradMatch) [19] match gradient norms via submodular maximisation; Error L2 Norm (EL2N) and Gradient Normed (GraNd) [20] use early-training gradient norms. All of these methods were designed for and evaluated on differentiable models. Translating their selection signals to the GBDT setting is non-trivial because GBDTs are not differentiable with respect to the training set; we include adapted variants in our 17-method benchmark and show that their landscape fidelity, measured by SLD, predicts their tabular AUROC. The GBDT-native subsampling methods Gradient-based One-Side Sampling (GOSS) [9] and Minimal Variance Sampling (MVS) [21] operate per boosting round rather than producing a static condensed set; MVS is the theoretically closest prior work, as it explicitly minimises the variance of the split-scoring estimator. Data valuation [22], [23] and influence-function methods offer related but complementary perspectives on data contribution; they often require retraining on many subsets or differentiable-model assumptions and are not directly applicable in their standard form to histogram GBDTs.

c) Why GBDTs for tabular data.: Gradient-boosted decision trees [7]–[9], [24] remain one of the strongest practical model classes for tabular data [25], [26]. This motivates a condensation evaluation framework specific to tree-structured inductive bias. The split-gain criterion [27], [28] aggregates per-bin gradient statistics, and the resulting landscape fully determines which feature and threshold a node selects, a structure that has no direct analogue in differentiable models. Our SLD exploits this structure to build a measurement instrument that is GBDT-mechanism-specific: it explains *why* a condensed set preserves tree behaviour, not merely whether it does.

d) The C²TC contrast.: The most closely related recent work is C²TC (Class-Adaptive Clustering for Tabular Condensation) [6] (ICDE 2026), a tabular condensation method that optimises class allocation and feature representation through class-adaptive clustering. Our work is different in intent: C²TC is a condenser, while SLD is a diagnostic that audits whether any surrogate preserves the GBDT split-gain landscape under a fixed prediction state. We did not include C²TC in the 17-method comparative grid because its official implementation was not directly compatible with our frozen weighted-surrogate benchmarking interface at the time of grid construction. To avoid leaving this interaction untested, we ran a compatibility audit using the official C²TC code on the Adult split at budget 200. The generated surrogate achieved AUROC 0.769, but had raw $SLD_{\infty} = 1228.21$, relative

$SLD_\infty = 4.72$, and root agreement = 0.0. We interpret this as a compatibility and scope check, not a comparative result: budget 200 and the synthetic C²TC surrogate fall in exactly the regime where our own scale and margin analyses warn that root-level SLD can lose fine AUROC resolution. The run confirms that SLD can audit C²TC-generated surrogates, but it is not a full C²TC baseline.

III. BACKGROUND AND THE SLD DIAGNOSTIC

A. GBDT Split-Gain Landscape

Let the full training dataset be $D = \{(x_i, y_i)\}_{i=1}^N$ with features $x_i \in \mathbb{R}^F$ and labels y_i . A gradient-boosted decision tree (GBDT) fits an additive ensemble of trees by performing, at each node, a greedy split that maximises the regularised gain [7], [8]. Given first- and second-order loss statistics $g_i = \partial_{\hat{y}} \ell(\hat{y}_i, y_i)$ and $h_i = \partial_{\hat{y}}^2 \ell(\hat{y}_i, y_i)$ for each instance i , the gain of partitioning the instance set I at a node into left and right children I_L, I_R is

$$\mathcal{G}(I_L, I_R) = \frac{1}{2} \left[\frac{(\sum_{i \in I_L} g_i)^2}{\sum_{i \in I_L} h_i + \lambda} + \frac{(\sum_{i \in I_R} g_i)^2}{\sum_{i \in I_R} h_i + \lambda} - \frac{(\sum_{i \in I} g_i)^2}{\sum_{i \in I} h_i + \lambda} \right] - \gamma, \quad (1)$$

where $\lambda \geq 0$ is the ℓ_2 leaf-weight regulariser and $\gamma \geq 0$ is the minimum gain threshold [8]. Modern histogram-based GBDTs [9], [24] discretise each feature f into B bins; per-bin sums of (g, h) suffice to evaluate every candidate split.

Lemma 1 (Histogram sufficiency; standard, e.g. [9], [21]). *Given the current gradients and Hessians, a histogram GBDT with B bins per feature has its full per-node split-gain landscape determined entirely by the $F \times B$ grid of per-bin (g, h) aggregates. No other information from the training set is required to evaluate (1) at every candidate (feature, threshold) pair.*

We define the *split-gain landscape* $\Gamma_D \in \mathbb{R}^{F \times (B-1)}$ as the gain $\mathcal{G}$ that each (feature, bin-boundary threshold) candidate would achieve on dataset D under a fixed prediction state. Lemma 1 is a standard consequence of the histogram GBDT algorithm; we state it here to ground the diagnostic. Its importance is histogram sufficiency: once the prediction state fixes g_i, h_i , Γ_D can be computed in one histogram pass.

B. Split-Landscape Discrepancy

Let $D' = \{(x'_j, y'_j, w'_j)\}_{j=1}^m$ be a weighted selected or synthetic surrogate of D , with $m \ll N$. Synthetic points are binned using the same feature preprocessing and bin boundaries as D ; their labels may be hard labels or soft labels converted to the objective's expected gradients. We quantify the structural gap between D and D' via their landscape distance.

Definition 1 (Split-Landscape Discrepancy). *Let the raw landscape error be $\delta_p(D, D') = \|\Gamma_D - \Gamma_{D'}\|_p$. The relative*

Split-Landscape Discrepancy of order p between dataset D and surrogate D' is

$$SLD_p(D, D') = \frac{\delta_p(D, D')}{\max(\|\Gamma_D\|_p, \epsilon)}, \quad (2)$$

using the same bins, objective, regularisation, base score, and feature preprocessing for D and D' . We use $p = \infty$ (worst-case relative split distortion) and the rank-based split-regret variant. Root- SLD compares the root landscape. Reference-tree early-node SLD averages over early nodes of a fixed full-data reference tree, using the reference node memberships. Full-path SLD extends the same comparison along deeper reference-tree paths.

Both variants are computed from histogram statistics under a fixed prediction state, and crucially *no model is retrained on D'* . Figure 1 illustrates how SLD fits into the condensation evaluation pipeline as a no-candidate-retraining alternative to the retrain-and-evaluate loop.

C. Diagnostic Sanity Checks

The diagnostic rests on two standard facts about histogram GBDTs and set coverage. We use them only to motivate what SLD measures; the contribution of the paper is the empirical measurement study across 17 condensation methods.

a) *Margin preservation.*: Let Δ_v be the gain margin between the best-ranked and second-best-ranked split at node v on D . If

$$\delta_\infty(D, D') < \Delta_v/2, \quad (3)$$

then the argmax split selected by D' coincides with the argmax selected by D at node v , assuming a unique argmax and deterministic tie-breaking. In the dose-response cells satisfying this raw-error margin condition, root-level split agreement equals 1.0 (Section V-C).

b) *Coverage guarantee.*: The split-coverage objective is the unweighted fraction of full-data high-gain split thresholds, the top- k full-data split candidates used by the coverage selector, whose adjacent bins are represented by at least one selected or weighted surrogate point. It is binary threshold coverage, not gain-weighted path fidelity. As a standard set-coverage objective, it is monotone submodular under a cardinality constraint, so greedy maximisation has the usual $(1 - 1/e)$ guarantee [29]. This defines HistDistill-Greedy: it covers the split landscape near-optimally yet performs poorly by downstream predictive performance (Section V-B). The gap between coverage and predictive fidelity is the central empirical finding of Section V-B.

D. Computing SLD: Algorithm

Algorithm 1 summarises the SLD computation. The key property is that, after the fixed prediction state is available, scoring a candidate requires only one histogram pass and one arithmetic comparison of landscapes; no candidate GBDT is trained. After shared binning, histogram

Algorithm 1 Compute Split-Landscape Discrepancy without candidate retraining

Require: Full dataset D , weighted surrogate D' , bins, norm order p , fixed prediction state π

Ensure: $SLD_p(D, D')$ and split-regret score

- 1: Compute first- and second-order loss statistics for D under π $\triangleright$ root base score or reference-boosters predictions
- 2: Build histogram $H_D \in \mathbb{R}^{F \times B \times 2}$: for each feature f and bin b , store $(\sum_{i: x_{if} \in b} g_i, \sum_{i: x_{if} \in b} h_i)$
- 3: Compute landscape $\Gamma_D \in \mathbb{R}^{F \times (B-1)}$ via (1) applied to every candidate split
- 4: Repeat Steps 1–3 for D' to obtain $\Gamma_{D'}$ $\triangleright$ selected rows reuse weights; synthetic rows are evaluated under π
- 5: $SLD_p \leftarrow \|\Gamma_D - \Gamma_{D'}\|_p / \max(\|\Gamma_D\|_p, \epsilon)$
- 6: **Split-regret:** let $(f^*, b^*) = \arg \max_{f,b} [\Gamma_D]_{f,b}$ and $(\hat{f}, \hat{b}) = \arg \max_{f,b} [\Gamma_{D'}]_{f,b}$;
- 7: split-regret $\leftarrow \max\left(0, \frac{[\Gamma_D]_{f^*, b^*} - [\Gamma_{D'}]_{\hat{f}, \hat{b}}}{\max(\|\Gamma_D\|_p, \epsilon)}\right)$
- 8: **return** SLD_p , split-regret

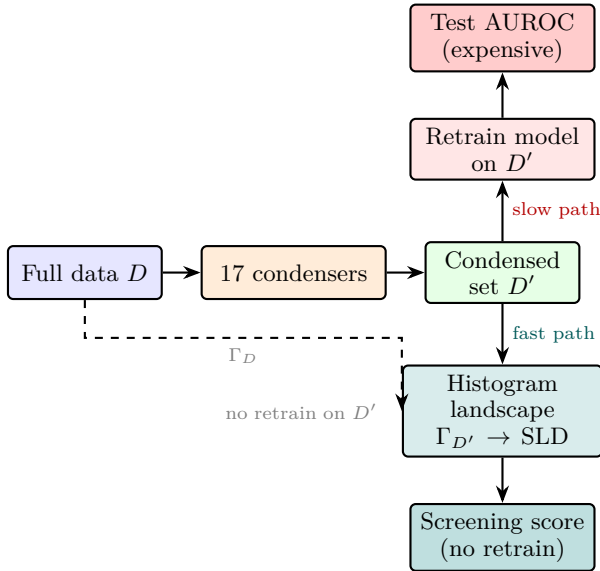


Fig. 1: **SLD evaluation pipeline.** After selecting a condensed set D' from full data D via any of 17 condensers, the fast path computes the candidate’s histogram split-gain landscape $\Gamma_{D'}$ under the same fixed prediction state as Γ_D . It provides a cheap screening signal before retraining on each candidate, in contrast to the slow path of retrain-and-test evaluation.

accumulation costs $O(NF)$ for full data and $O(mF)$ for the surrogate; prefix-sum gain evaluation and memory are $O(FB)$ per node.

IV. EXPERIMENTAL SETUP

A. Methods Evaluated

We evaluate 17 condensation methods spanning five groups; Table I gives the full taxonomy. **Geometry/classical coresets** include Random sampling, Herd-

ing [3], and K-center [18]. **GBDT-native subsampling** methods include GOSS [9] and MVS [21], adapted to produce static condensed sets. **Gradient- and score-based selection** methods include GraNd, EL2N [20], CRAIG-style [4], GradMatch-style [19], and gradient sampling (loss-weighted). **Distribution-matching distillation** is represented by DistributionMatching [10] adapted to tabular features. **Gain/landscape-based methods** form our primary family: HistDistill-Greedy (the $(1 - 1/e)$ -optimal coverage selector), HistDistill-Refined and HistDistill-Density (controlled landscape-fidelity diagnostic baselines), gain sketch, gain-path sketch, and the legacy gain-path refined method. The last three are prior gain-path methods; the paper’s main claim is the diagnostic and benchmark, not a new state-of-the-art condenser. All adaptations produce static surrogates: GOSS/MVS select the top scored instances under the fixed GBDT state; GraNd, EL2N, CRAIG, and GradMatch use their published scoring or matching rules; DistributionMatching runs on processed tabular features and shares full-data bin edges for SLD scoring. Short names follow the original methods: GOSS is Gradient-based One-Side Sampling, MVS is Minimal Variance Sampling, GraNd is Gradient Normed, EL2N is Error L2 Norm, CRAIG is a gradient-matching coreset method, DM abbreviates Distribution-Matching, and HD abbreviates HistDistill.

B. Datasets and Protocol

Our experiments use public tabular datasets drawn from OpenML [30] and the Penn Machine Learning Benchmarks (PMLB) [31]; several entries originate from UCI-style benchmark datasets. The full inventory appears in Table X. The repository lists the exact OpenML/PMLB identifiers and processed split checksums. The core study is on binary classification. For each dataset we condense the training set to five budgets, $\{10, 25, 50, 100, 200\}$ rows, and repeat every (dataset, method, budget) configuration over five random seeds. Across the 26 binary datasets and the five core methods this gives 3,250 retrained-model evaluations for headline performance (Table II) and 650 landscape-fidelity measurements, one per dataset, method, and budget.

The broader 17-method audit uses the same 26 datasets, budgets 25/50, and three seeds, giving 156 cells per method for Figure 2. The decomposition table is a diagnostic subset used only to illustrate the two reference-chain failure modes, so its absolute AUROCs are not compared to the pooled headline values. Additional probes include the 1,872-cell split-noise robustness study, multiclass, regression, hyperparameter-optimization (HPO) transfer, and a Coverttype/SUSY/HIGGS scale audit with 400-row condensed sets; Sections V-E and VI give details.

C. Metrics

Downstream predictive performance. We score binary tasks by area under the receiver operating charac-

teristic curve (AUROC), a threshold-free ranking metric; multiclass tasks by one-vs-rest macro-AUROC, the class-averaged AUROC; and regression tasks by coefficient of determination (R^2). Every score is computed on a held-out test set with the model retrained on the condensed set. Reported 95% CIs are confidence intervals over dataset clusters unless a table states otherwise.

Fidelity. (2) defines the relative ℓ_∞ landscape distance, while split-regret is the normalised gain lost when a node follows the surrogate’s argmax split rather than the full-data one. The reported SLD columns and Figure 3 use raw ℓ_∞ landscape error to expose the scale confounding in Finding 2; these raw values are not cross-dataset scores. Split-regret is the primary normalised ranking metric. Both are read off the histogram statistics before retraining on any candidate surrogate.

Error decomposition. To separate the two ways a condensed set can hurt a tree, we evaluate three models in a reference chain. The *structure reference* uses the full-data tree structure with leaves refit under the decomposition protocol. The *leaf-refit model* uses the condensed-set tree structure but refits its leaves on full data, isolating the cost of the structure alone. The *deployed model* is trained only from the condensed set. We define structure error as reference AUROC minus leaf-refit AUROC, and leaf-estimate gap as leaf-refit AUROC minus deployed AUROC. These two gaps add along the reference-to-deployed chain; they are not intended to sum to the ordinary full-data AUROC gap. We also track how representatively the condensed set fills the model’s leaves, defined as the ℓ_1 distance between the condensed- and full-data distributions of training instances across the leaves; we call this the leaf-population mismatch.

D. Learners and Reproducibility

The reference landscape and primary downstream learner use histogram XGBoost. Robustness checks additionally use LightGBM, CatBoost/fallback histogram boosting, random forests, and multilayer perceptrons (MLPs). All datasets are public and preprocessing is deterministic: categorical features follow the repository scripts, numerical features use shared full-data bin edges, and splits use stratified seeds where labels permit. The repository, <https://github.com/VIVAN-RANGRA/sld-gbdt-condensation>, contains the exact hyperparameters, seeds, package versions, processed metric CSVs, plotting code, and commands needed to reproduce the reported tables.

V. RESULTS

A. Does the Landscape Predict Performance?

We first observe in Figure 2 and Table III that method-level split-regret ranks method families in the small-to-mid-size benchmark. The score is computed before candidate retraining and is strongly monotone with eventual AUROC at the method-mean level: gain-path methods sit

TABLE I: The 17 condensation methods. †: adapted from differentiable settings.

Family	Method	Signal / citation
Geometry	Random	Uniform Moments [3] Coverage [18]
	Herding K-center	
GBDT-native	GOSS†	Grad-norm [9]
	MVS†	Min-var [21]
Gradient	GraNd†	Norm at init [20]
	EL2N†	Error L2 [20]
	CRAIG†	Submod. [4]
	GradMatch†	Grad match [19]
	Grad. samp.†	Loss-weighted
Distrib.	DistributionMatching	Distr. match [10]
Gain / landscape	HD-Greedy	(1 - 1/e) coverage
	HD-Refined	Landscape fidelity
	HD-Density	Land.+density
	Gain sketch	Gain sketch
	Gain-path sk.	Gain-path sketch
	Legacy gp	Gain-path refined

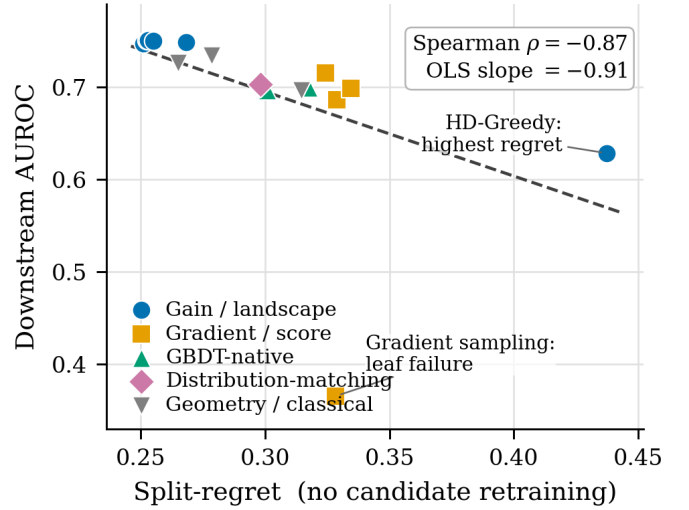


Fig. 2: **No-candidate-retraining method audit.** Each point is one condensation method. Lower split-regret, computed before candidate retraining, tracks higher downstream AUROC over 156 dataset–budget–seed cells (Spearman $\rho = -0.87$, OLS slope -0.91).

in the low-regret/high-AUROC corner, while the coverage-optimal HD-Greedy method sits at the opposite extreme. Gradient sampling is the main exception, and the decomposition below shows why: its problem is leaf-value estimation, not split-structure preservation. This 17-method diagnostic grid uses 26 datasets, two audited budgets, and three seeds, and is separate from the five-budget, five-seed core grid in Table II. The large-scale audit in Section V-E bounds the claim: once high-gain margins are cleared on very large datasets, root split-regret loses rank resolution.

The gradient-sampling AUROC is an audited failure mode rather than a plotting artifact: its deployed AUROC is also low in the reference-chain subset (Table V), where the error is leaf-estimate dominated. The method-level association is not driven only by this point or by HD-Greedy: excluding both leaves Spearman $\rho = -0.85$ over the remaining 15 methods.

TABLE II: Headline performance and diagnostics for five core methods (3,250 cells). AUROC uses mean $\pm$ 95% CI over dataset clusters; avg. rank lower is better. Raw SLD $_{\infty}$ is diagnostic only and is not a cross-dataset fidelity score; speedup is relative to full-data retraining on the same dataset/learner.

Method	AUROC	Rank	Raw SLD $_{\infty}$ diag.	Speedup
Full data	0.812 $\pm$ 0.062	—	—	1.0 $\times$
Legacy gain-path	0.743 $\pm$ 0.057	2.60	99.4	3.9 $\times$
HistDistill-Refined	0.742 $\pm$ 0.057	2.68	93.6	3.5 $\times$
HistDistill-Density	0.742 $\pm$ 0.057	2.65	93.7	3.5 $\times$
Herdling	0.731 $\pm$ 0.054	3.44	174.1	3.8 $\times$
Random	0.724 $\pm$ 0.056	3.63	138.3	4.0 $\times$

TABLE III: Seventeen-method diagnostic audit. Values are method means $\pm$ 95% CI over dataset clusters, over 156 dataset-budget-seed cells. Lower split-regret and higher AUROC are better. EL2N and GraNd coincide after rounding under the fixed-state adaptation used here.

Method	AUROC	Regret
Gain-path sketch	0.748 $\pm$ 0.062	0.251 $\pm$ 0.079
Legacy gain-path	0.751 $\pm$ 0.062	0.252 $\pm$ 0.079
HD-Density	0.751 $\pm$ 0.061	0.253 $\pm$ 0.073
HD-Refined	0.750 $\pm$ 0.062	0.255 $\pm$ 0.077
Random	0.726 $\pm$ 0.060	0.265 $\pm$ 0.064
Gain sketch	0.749 $\pm$ 0.062	0.268 $\pm$ 0.081
Herdling	0.734 $\pm$ 0.059	0.279 $\pm$ 0.082
DM	0.703 $\pm$ 0.062	0.298 $\pm$ 0.075
MVS	0.697 $\pm$ 0.058	0.301 $\pm$ 0.068
K-center	0.697 $\pm$ 0.062	0.315 $\pm$ 0.079
GOSS	0.700 $\pm$ 0.058	0.317 $\pm$ 0.073
GradMatch	0.715 $\pm$ 0.057	0.324 $\pm$ 0.086
Grad. samp.	0.366 $\pm$ 0.067	0.328 $\pm$ 0.078
EL2N	0.686 $\pm$ 0.056	0.329 $\pm$ 0.071
GraNd	0.686 $\pm$ 0.056	0.329 $\pm$ 0.071
CRAIG	0.699 $\pm$ 0.061	0.335 $\pm$ 0.082
HD-Greedy	0.629 $\pm$ 0.050	0.437 $\pm$ 0.106

Negative leaf gaps mean deployed condensed leaves slightly outperform the full-data leaf refit under this diagnostic chain, usually through regularisation or favourable condensed weighting.

We also observe in Figure 3 that lower raw SLD predicts higher AUROC within a dataset, but pooling across datasets is confounded. Within each dataset, lower raw δ_{∞} corresponds to better retrained AUROC; after pooling datasets, the trend can invert because dataset scale affects both raw SLD magnitude and absolute AUROC. Raw SLD should therefore be used for within-dataset, within-budget method comparison, not as a cross-dataset absolute score.

B. Coverage Is Not Predictive Fidelity

Figure 4 and Table IV show that the near-optimal coverage selector is weak by downstream predictive performance. HD-Greedy optimises split-threshold coverage and achieves the expected coverage win, but this objective is not the one that preserves downstream AUROC. The method lands in the bottom tier of the 17-method audit (Table III); unweighted threshold coverage and gain-weighted landscape fidelity are different objectives.

Figure 5 and Table V further show that the coverage failure is structural, whereas gradient sampling fails for a different reason. The reference chain separates structure error from leaf-estimate gap. HD-Greedy’s loss comes from choosing the wrong gain-weighted structure, not from poor

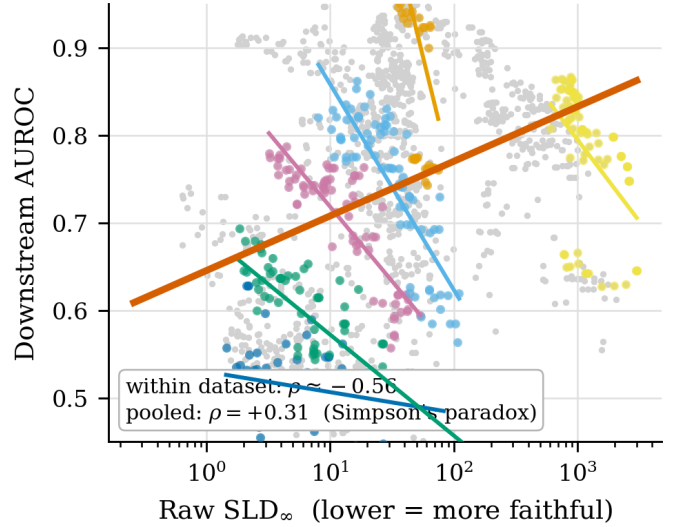


Fig. 3: **Within-dataset SLD trend.** Coloured lines are within-dataset fits on log-scale raw SLD; the average within-dataset trend is negative. The pooled fit reverses because dataset scale confounds raw SLD and AUROC.

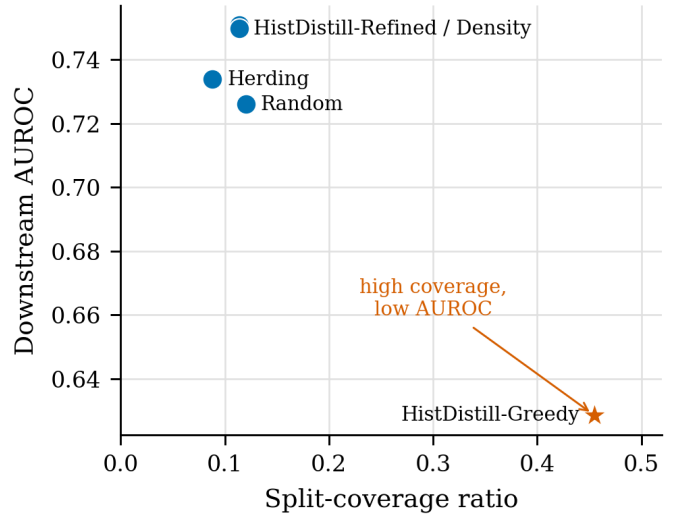


Fig. 4: **Coverage is not predictive fidelity.** HD-Greedy covers the most high-gain split thresholds, but the accurate methods occupy a different region of the coverage–AUROC plane.

TABLE IV: Coverage metrics versus AUROC on the two-budget coverage audit. Cov. is high-gain split coverage; Proxy is coverage divided by the proxy upper bound.

Method	Cov25	Proxy25	Cov50	Proxy50	AUROC
HD-Greedy	0.371	1.000	0.539	1.000	0.629
Random	0.082	0.246	0.159	0.299	0.726
HD-Refined	0.075	0.223	0.152	0.286	0.750
HD-Density	0.075	0.223	0.152	0.286	0.751
Herdling	0.059	0.183	0.118	0.229	0.734

leaf values after its structure is fixed. Gradient sampling is the opposite diagnostic case: its population mismatch is

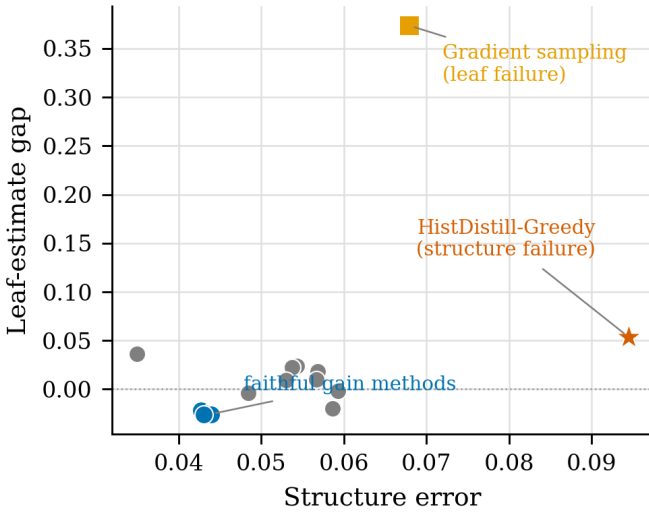


Fig. 5: **Two failure modes.** HD-Greedy is structure-dominated; gradient sampling is leaf-estimate dominated. Faithful gain methods occupy the low-error corner.

TABLE V: Error decomposition on the diagnostic subset used to illustrate both failure modes. Absolute AUROCs differ from the pooled values in Table II and Figure 2. Structure error is reference minus leaf-refit; leaf gap is leaf-refit minus deployed.

Method	Dep.	Leaf	Str. Err	Leaf Gap	Pop.
HD-Greedy	0.643	0.696	0.094	0.054	0.243
Grad. samp.	0.349	0.723	0.068	0.374	0.133
DM	0.720	0.756	0.035	0.036	0.235
K-center	0.713	0.737	0.054	0.024	0.196
MVS	0.724	0.734	0.057	0.010	0.088
GradMatch	0.734	0.732	0.059	-0.002	0.113
HD-Refined	0.773	0.747	0.044	-0.026	0.060
Legacy gp	0.774	0.748	0.043	-0.026	0.068

Decomposition-subset anchors: full AUROC = 0.826, reference = 0.791.

low, but its leaf-estimate gap is high, pointing to distorted within-leaf gradient/Hessian or reweighting statistics.

C. How Much Can Structure Be Perturbed?

Figure 6 and Table VI show a margin-linked robustness pattern for GBDT structure by injecting calibrated split-gain noise into the full-data landscape. AUROC and split agreement stay flat while the margin condition in (3) holds; beyond that point, agreement drops first and AUROC degrades mildly. The result explains why aggressive condensation can work when the dominant split margins are preserved, and why corrupting high-gain structure is disproportionately costly.

D. Performance, Fidelity, and Cost

Tables II and IX show practical parity at the top of the core grid: HistDistill-Refined and HistDistill-Density match the legacy gain-path selector in AUROC while achieving the lowest landscape discrepancy among the core methods. The Refined-vs.-Legacy mean difference is operationally negligible and its dataset-clustered confidence interval crosses zero. At the studied small-data budgets,

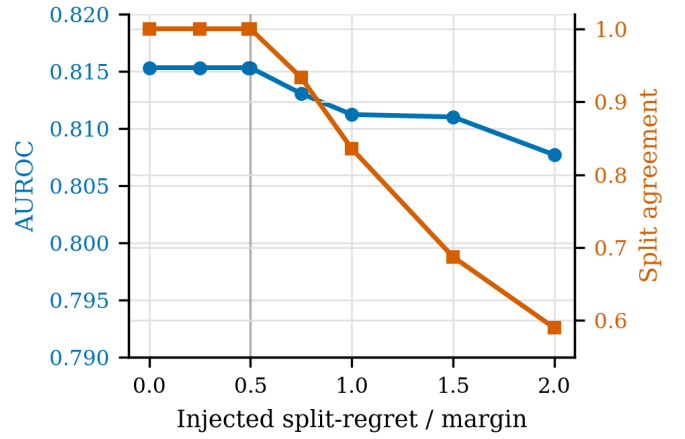


Fig. 6: **Robustness pattern.** Split agreement drops after dose 0.5; AUROC degrades mildly as calibrated split-gain noise exceeds the margin.

TABLE VI: Dose-response: 26 binary datasets, 1,872 cells. Values are mean $\pm$ 95% CI over dataset clusters; eight injected-noise levels are grouped into five displayed rows.

Dose	AUROC	Δ AUROC	Agree.	Regret
≤ 0.50	0.815 ± 0.068	0.000	1.000 ± 0.000	0.000
0.75	0.813 ± 0.069	-0.002	0.933 ± 0.002	0.011
1.00	0.811 ± 0.069	-0.004	0.836 ± 0.004	0.030
1.50	0.811 ± 0.069	-0.004	0.687 ± 0.008	0.069
2.00	0.808 ± 0.070	-0.008	0.590 ± 0.011	0.101

condensed retraining gives a median 3–4 $\times$ speedup, with median condensed training under half a second in the recorded runs; SLD itself costs one histogram pass.

E. Does the Diagnostic Hold at Scale?

Scale audit: the cost payoff grows with N , while ordering is dataset-dependent. Figure 7 and Tables VII and VIII report a real large-data audit using 400-row condensed sets. Speedup rises from single digits at 20k rows to hundreds or thousands at million-row scale. Random has the best large- N AUROC point estimate on all three datasets, but the three-dataset paired audit is underpowered (two-sided Wilcoxon $p = 0.25$ vs. HD-Refined and HD-Density), so the practical reading is convergence among reasonable condensers. This is the regime thesis: HD-Refined beats Random in 24/26 small-to-mid-size datasets (Table IX), yet Random has the largest point estimate on all three very large datasets (Table VIII).

HIGGS illustrates the same margin mechanism as Figures 3 and 6: several methods have zero root split-regret because the dominant root split is saturated, so leaf estimates or deeper structure determine AUROC. Root-level regret still rejects severe failures, but it cannot finely rank reasonable surrogates once the top split margin is cleared.

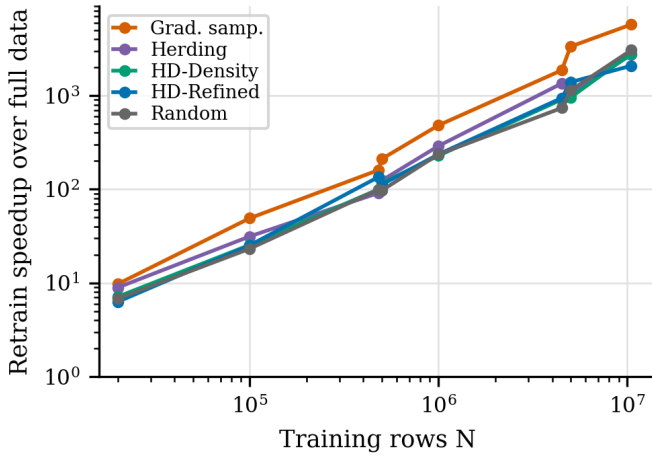


Fig. 7: **Large-scale retraining payoff.** Log-log speedup for 400-row condensed retraining on Covertypes, SUSY, and HIGGS. Points are single runs at observed dataset sizes, so no run-to-run CI is claimed for this scale audit.

TABLE VII: Large-scale audit at the largest training size per dataset. ρ is Spearman correlation over the five audited methods; negative ρ means lower split-regret aligns with higher AUROC. Each row is a single large-scale run.

Dataset	Rows	ρ	Top speedup
Covertypes	0.48M	-0.36	160×
SUSY	4.50M	-0.97	1,864×
HIGGS	10.50M	0.11	5,734×

TABLE VIII: Large-scale AUROC at the largest training size. Full-data AUROCs are 0.897, 0.873, and 0.803 for Covertypes, SUSY, and HIGGS. Random wins the condensed point estimates, but the three-dataset paired audit is underpowered ($p = 0.25$ against HD-Refined and HD-Density); no per-row CI is claimed because the large-scale audit uses one full run per dataset/method.

Method	Cov.	SUSY	HIGGS
Full data	0.897	0.873	0.803
Random	0.792	0.839	0.687
HD-Refined	0.747	0.818	0.648
HD-Density	0.781	0.794	0.641
Herding	0.719	0.780	0.450
Grad. samp.	0.377	0.219	0.364

TABLE IX: Dataset-clustered core-method comparisons. Δ is the paired mean AUROC difference over 650 dataset-budget-seed cells; confidence intervals bootstrap the 26 dataset clusters.

Method	Comparator	Δ	95% CI	DS wins
HD-Refined	Legacy gp	-0.00056	[-0.0019, 0.0008]	11/26
HD-Density	Legacy gp	-0.00056	[-0.0016, 0.0005]	9/26
HD-Refined	Herding	0.01183	[0.0037, 0.0204]	18/26
HD-Refined	Random	0.01807	[0.0124, 0.0233]	24/26

VI. WHERE THE LENS STOPS: SCOPE AND LIMITATIONS

The SLD diagnostic is designed around the inductive bias of histogram GBDTs. Three scopes delimit where that

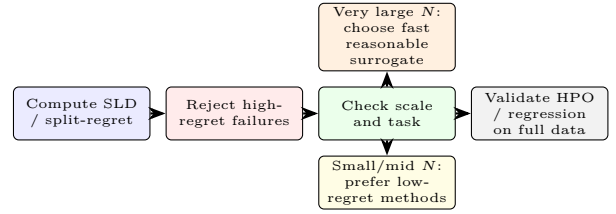


Fig. 8: **Operational workflow implied by the measurements.** SLD is most useful as a cheap screening diagnostic: reject high-regret failures, prefer low-regret methods only when margins are not saturated, and validate HPO, regression, and final deployment choices on full data.

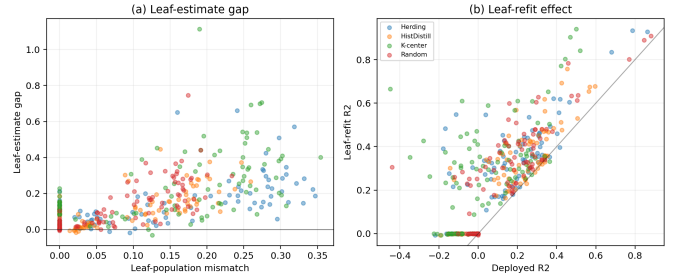


Fig. 9: **Regression failure modes.** Left: leaf-population mismatch tracks leaf-estimate loss. Right: refitting leaves on full data helps, but the remaining structure gap dominates.

design leads to weak or negative results.

Regression is structure-dominated, and no condenser fixes it. The regression audit uses eight public datasets (diabetes, California housing, abalone, cpu_act, kin8nm, house_16H, elevators, wine quality), budgets 25/50, seeds 0–2, and XGBoost/LightGBM/MLP downstream learners. SLD is recomputed for squared-error gradients. The structure reference—the full-data tree structure with leaves refit under the decomposition protocol—achieves $R^2 \approx 0.63$. The leaf-refit model, which uses each condensed set’s structure but refits leaves on full data, falls to $R^2 = 0.28$ – 0.29 . The deployed condensed models then reach only $R^2 \approx 0.16$ for HistDistill and 0.05 for K-center. Thus, for HistDistill, the total gap from the structure reference to the deployed model is about 0.47; of that gap, about 0.35 is structural ($0.63 - 0.28$) and about 0.12 is due to leaf-value estimation ($0.28 - 0.16$). No method we evaluate closes the structure gap: the best deployed condensed R^2 is 0.16, well below even the leaf-refit baseline of 0.28. Figure 9 shows this pattern. The SLD diagnostic correctly *diagnoses* this failure, since the condensed sets have high SLD_∞ relative to full-data regression baselines, but it does not cure it. Regression tasks appear to require structure fidelity that current condensers cannot achieve at the budgets we study.

HPO transfer is weak. We test whether condensed sets preserve hyperparameter rankings on 12 bi-

nary datasets using HistDistill-Refined at budget 50. Each dataset evaluates the same 100 XGBoost hist configurations on the condensed and full train/validation splits; the random-search space is 50–150 trees, depth 2–5, learning rate 0.025–0.18, min-child-weight 0.5–8, subsample/column-subsample 0.65–1, $\lambda = 0.05$ –8, and $\gamma = 0$ –2. Across datasets, top-3 Jaccard between condensed- and full-data HPO results is 0.042, top-1 match rate is 0.083, and Spearman rank correlation is 0.145. Median HPO speedup is $1.57\times$. The mean final test-AUROC gap is only 0.00125, but that is because condensed HPO finds a different competitive configuration, not because it transfers the full-data ranking. Condensed HPO should therefore be followed by full-data checks of top configurations before deployment.

No-candidate-retraining selection of the single best method remains weak. SLD ranks method families well at the aggregate level (Spearman -0.87), but selecting the *single best* method for a new dataset based only on SLD is harder. The top-3 success rate of the SLD-based selector is 0.436 across 156 dataset–budget–seed cells, and the top-1 rate is 0.122. The split-regret selector performs similarly. The landscape score is useful for screening out bad condensers and ranking families, but the gap between the top-2 or top-3 methods is often smaller than the within-dataset variance of SLD, making single-method selection unreliable. Future work could combine the landscape score with dataset-level meta-features to improve per-instance selection.

Across eight multiclass datasets, HistDistill obtains 0.910 one-vs-rest macro-AUROC versus Random 0.897, a thin positive extension rather than a main claim; overall, SLD is built around the GBDT split-gain objective, and its predictive value erodes as a task drifts away from that inductive bias.

VII. CONCLUSION

Measuring mechanism, not just downstream predictive performance, reveals things that a retrain-and-compare protocol cannot. For gradient-boosted decision trees on tabular data, the split-gain landscape is a central mechanism for tree-structure selection: it determines which features and thresholds the booster commits to before a leaf value is estimated. Comparing landscapes between a full dataset and its surrogate, through the Split-Landscape Discrepancy, yields a no-candidate-retraining diagnostic whose method-level split-regret correlates with downstream AUROC in the small-to-mid-size 17-method diagnostic grid at Spearman -0.87 . The scale audit qualifies this claim: once very large datasets make the dominant split margins easy to preserve, random sampling becomes difficult to beat and root split-regret becomes a rejection filter rather than a fine-grained ranker.

The most practically consequential finding is that *coverage is not predictive fidelity*. A condensation method with

a provable near-optimal guarantee on unweighted split-threshold coverage is simultaneously among the weakest downstream performers in the pooled audit. The reason is structural: unweighted coverage fills the landscape broadly, while AUROC depends on gain-weighted, Hessian-weighted, path-sensitive structure preservation. The reference-chain decomposition makes this precise: the coverage-optimal method’s failure is a structure failure rather than a leaf-value failure.

We also delimit the scope of these findings. Regression tasks are structure-dominated in a way that current condensers cannot overcome. Hyperparameter transfer from condensed sets to full-data rankings is weak. At deployment scale, reasonable condensers can converge in root split-regret while still differing in leaf estimates and deeper structure. These boundaries reinforce the role of SLD as a rejection filter and measurement instrument, not a single-best-method selector.

The applied workflow is therefore direct: before choosing a condensation strategy, compute SLD in a histogram pass and use split-regret to reject bad candidates. This screens out poor surrogates before retraining on every candidate; when N is very large and several candidates clear the dominant split margins, choose among the reasonable surrogates by budget, speed, and downstream validation. Coverage metrics should not be used as an AUROC proxy, and for regression tasks or HPO transfer condensed sets should be treated as approximations to validate rather than substitutes for the full data. We release the diagnostic and benchmark in the repository to make this workflow reproducible and extensible to new condensation methods as they emerge.

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Appendix: Dataset Inventory

TABLE X: Expanded 26-dataset binary benchmark inventory. OpenML entries give dataset ID/task ID when a task was used; PMLB entries use exact PMLB identifiers. *: focused binary suite. N and F are reported after target removal, benchmark preprocessing, and the 20,000-row cap where applied.

Dataset	ID	N	F	Dataset	ID	N	F
adult*	PMLB:adult	20,000	14	gametes_epi_1000	PMLB:GAMETES_Epistasis_2_Way_1000atts_0.4H_EDM_1_EDM_1_1	1,600	1000
agaricus_lepiota	PMLB:agaricus_lepiota	8,145	22	gametes_epi_20_0.1H	PMLB:GAMETES_Epistasis_2_Way_20atts_0.1H_EDM_1_1	1,600	20
australian*	OpenML d40981	690	14	gametes_epi_20_0.4H	PMLB:GAMETES_Epistasis_2_Way_20atts_0.4H_EDM_1_1	1,600	20
bank_marketing*	OpenML d1461	20,000	16	gametes_epi_3way	PMLB:GAMETES_Epistasis_3_Way_20atts_0.2H_EDM_1_1	1,600	20
breast_w*	OpenML d15/t15	699	9	gametes_het_50	PMLB:GAMETES_Heterogeneity_20atts_1600_Het_0.4_0.2_50_EDM_2_001	1,600	20
credit_approval	OpenML d29/t29	690	15	gametes_het_75	PMLB:GAMETES_Heterogeneity_20atts_1600_Het_0.4_0.2_75_EDM_2_001	1,600	20
credit_g*	OpenML d31/t31	1,000	20	hill_valley_with_noise	PMLB:Hill_Valley_with_noise	1,212	100
diabetes*	OpenML d37/t37	768	8	hill_valley_without_noise	PMLB:Hill_Valley_without_noise	1,212	100
electricity	OpenML d151/t219	20,000	8	kr_vs_kp	OpenML d3/t3	3,196	36
pc1*	OpenML d1068	1,109	21	pc3	OpenML d1050/t3903	1,563	37
pc4	OpenML d1049/t3902	1,458	37	phoneme	OpenML d1489	5,404	5
qsar_biodeg	OpenML d1494	1,055	41	sick	OpenML d38/t3021	3,772	29
spambase*	OpenML d44/t43	4,601	57	tic_tac_toe	OpenML d50/t49	958	9